

Differential Drive Robot :-

Robot kinematics :-

(Rad/sec) Right wheel

Forward

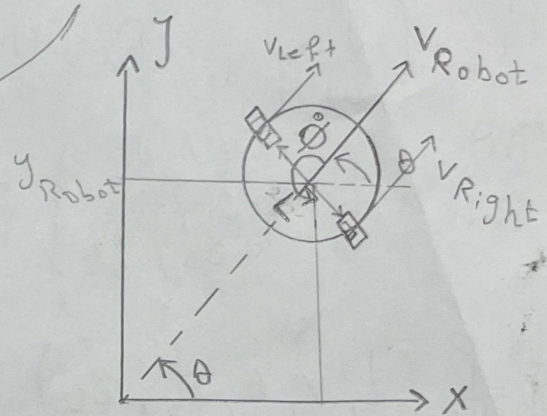
mobile Robot (m/sec)

(Rad/sec) Left wheel

Inverse

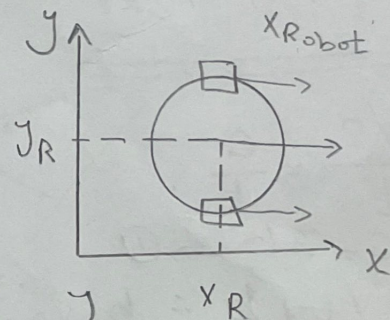
Position of the Robot :-

$$P = \begin{bmatrix} x_{Robot} \\ y_{Robot} \\ \theta \end{bmatrix}$$



[1] Translation motion at $\theta = 0$
"no Rotation"

$$\begin{bmatrix} T_x \\ T_y \end{bmatrix} = \begin{bmatrix} x_R \\ y_R \end{bmatrix}$$

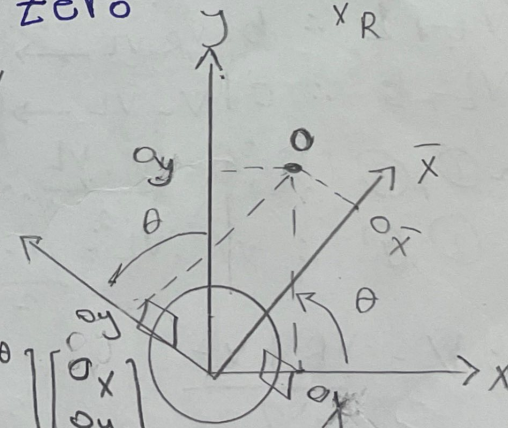


[2] Rotation at $x_R, y_R = \text{Zero}$
"no Translation"

$$O_x = O_{\bar{x}} \cos \theta - O_{\bar{y}} \sin \theta$$

$$O_y = O_{\bar{x}} \sin \theta + O_{\bar{y}} \cos \theta$$

$$\begin{bmatrix} O_x \\ O_y \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} O_{\bar{x}} \\ O_{\bar{y}} \end{bmatrix}$$



Rotation matrix about z

[3] General Case

$$\begin{bmatrix} \bar{x} \\ \bar{y} \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Differential Drive Robot Kinematics

$$P = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \text{ Position}$$

at time = 0

$$P_0 = \begin{bmatrix} x_0 \\ y_0 \\ \theta_0 \end{bmatrix}$$

at time = t

$$P_t = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix}$$

Position is change with time

$$\frac{\partial P}{\partial t} = \begin{bmatrix} \frac{\partial x_t}{\partial t} \\ \frac{\partial y_t}{\partial t} \\ \frac{\partial \theta_t}{\partial t} \end{bmatrix}$$

$$P^{\circ} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \text{Velocity}$$

$$\text{Velocity} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} V \\ \omega \end{bmatrix}$$

$$P^{\circ} = f_n(L, v, \theta, \dot{\theta}_R, \dot{\theta}_L)$$

where: $V = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix}$ Linear velocity

$\omega = \dot{\theta}$ angular velocity

$$\frac{L/2}{L} = \frac{C}{b}$$

$$\therefore C = 2b \rightarrow (1)$$

$$\therefore V_R = V_L + b \therefore b = V_R - V_L \rightarrow (2)$$

$$\therefore V = V_L + C \therefore C = V - V_L \rightarrow (3)$$

$$\text{From (1), (3)} \therefore b = \frac{V - V_L}{2} \rightarrow (4)$$

$$\text{From (2), (4)} \therefore V_R - V_L = \frac{V - V_L}{2}$$

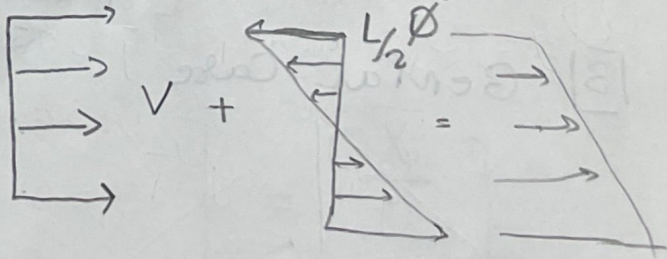
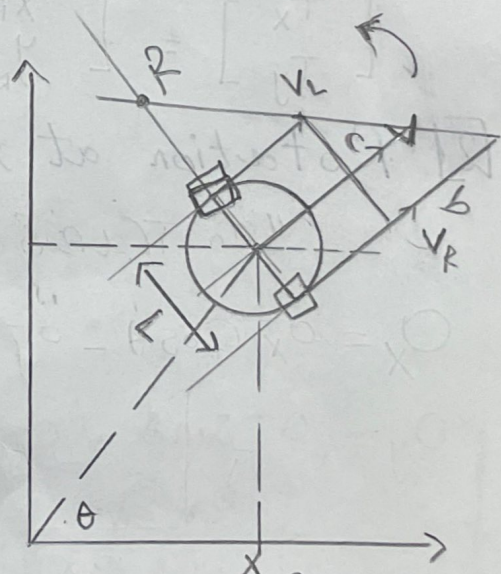
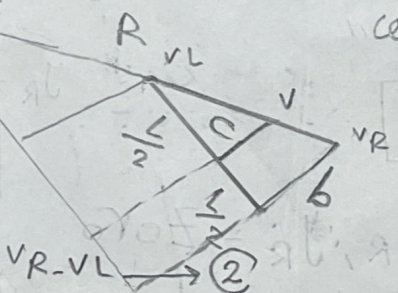
$$\therefore V = \frac{V_R + V_L}{2}$$

$$V_R = V + \frac{L}{2} \dot{\theta} \rightarrow (5)$$

$$V_L = V - \frac{L}{2} \dot{\theta} \rightarrow (6)$$

$\therefore$ From (5), (6)

$$V = \frac{V_R + V_L}{2}, \quad \omega = \dot{\theta} = \frac{V_R - V_L}{L}$$



$$\therefore V_R = \dot{\theta}_R \cdot r$$

$$\therefore V_L = \dot{\theta}_L \cdot r$$

$$\therefore V = \frac{\dot{\theta}_R + \dot{\theta}_L}{2} \cdot r \Rightarrow \text{Linear velocity}$$

$$\therefore \omega = \frac{\dot{\theta}_R - \dot{\theta}_L}{L} \cdot r \Rightarrow \text{angular velocity}$$

$$\begin{bmatrix} V \\ \omega \end{bmatrix} = \begin{bmatrix} \frac{r}{2} & \frac{r}{2} \\ \frac{r}{L} & -\frac{r}{L} \end{bmatrix} \begin{bmatrix} \dot{\theta}_R \\ \dot{\theta}_L \end{bmatrix} \rightarrow \text{①}$$

Robot velocity Conversion matrix wheels velocity

$$\begin{bmatrix} V \\ \omega \end{bmatrix} = M_s \begin{bmatrix} \dot{\theta}_R \\ \dot{\theta}_L \end{bmatrix}$$

$$\begin{bmatrix} \dot{\theta}_R \\ \dot{\theta}_L \end{bmatrix} = M_s^{-1} \begin{bmatrix} V \\ \omega \end{bmatrix}$$

$$\dot{P} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = R(\theta) \begin{bmatrix} V \\ \omega \end{bmatrix}$$

Rotat about z Cart Move in x axis

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V \\ \omega \end{bmatrix} \rightarrow \text{②}$$

from ①, ② $\Rightarrow$ DDJ-Forward kinematics

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{r}{2} & \frac{r}{2} \\ \frac{r}{L} & -\frac{r}{L} \end{bmatrix} \begin{bmatrix} \dot{\theta}_R \\ \dot{\theta}_L \end{bmatrix}$$

$$\therefore \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{r}{2} \cos \theta & \frac{r}{2} \cos \theta \\ \frac{r}{2} \sin \theta & \frac{r}{2} \sin \theta \\ \frac{r}{L} & -\frac{r}{L} \end{bmatrix} \begin{bmatrix} \dot{\theta}_R \\ \dot{\theta}_L \end{bmatrix} \quad \#$$

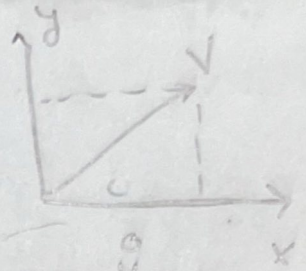
Jacobian

$$\dot{P} = J \cdot \dot{\theta}$$

DDR Inverse kinematics

$${}^0_0 \dot{P} = J \cdot \dot{Q}$$

$$\dot{Q} = J^{-1} \cdot \dot{P}$$



From: $V_R = \dot{\theta}_R \cdot r$, $V_L = \dot{\theta}_L \cdot r$

$$V_R = V + \frac{L}{2} \dot{\theta}, \quad V_L = V - \frac{L}{2} \dot{\theta}, \quad V = \dot{x} \cos \theta + \dot{y} \sin \theta$$

$$\dot{\theta}_R \cdot r = V + \frac{L}{2} \dot{\theta} \Rightarrow \dot{\theta}_R = [\dot{x} \cos \theta + \dot{y} \sin \theta + \frac{L}{2} \dot{\theta}] \cdot \frac{1}{r}$$

$$\dot{\theta}_L \cdot r = V - \frac{L}{2} \dot{\theta} \Rightarrow \dot{\theta}_L = [\dot{x} \cos \theta + \dot{y} \sin \theta - \frac{L}{2} \dot{\theta}] \cdot \frac{1}{r}$$

$$\begin{bmatrix} \dot{\theta}_R \\ \dot{\theta}_L \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \theta / r & \sin \theta / r & \frac{L}{2r} \\ \cos \theta / r & \sin \theta / r & -\frac{L}{2r} \end{bmatrix}}_{J^{-1}} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} \quad \#$$

Instantaneous radius:

$$R = \frac{V}{\dot{\theta}} = \frac{L}{2} \left| \frac{V_R + V_L}{V_R - V_L} \right|$$

Instantaneous coeff:

$$K = \frac{1}{R}$$

